MATHS MANTHAN
• TUITION CLASSES •
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**Subject:** Mathematics | **CBSE CLASS 10 – WEEKLY TEST 01** | **Time:** 45 Minutes
**Class:** Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 20**

**Student Name:** _____ **Roll No.:** _____ **Date:** _____

**GENERAL INSTRUCTIONS:**

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

## SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

**Q1.** The ratio of the length of a vertical rod and the length of its shadow on the ground is $1 : \sqrt{3}$. The angle of elevation of the sun is: [1]

(A) 30° (B) 45° (C) 60° (D) 90°

**Q2.** If the length of the shadow of a tower on the level ground is $\sqrt{3}$ times the height of the tower, then the angle of elevation of the sun is: [1]

(A) 45° (B) 30° (C) 60° (D) 90°

**Q3.** At a certain time of the day, if the length of the shadow of a vertical pole is equal to its height, then the sun's altitude is: [1]

(A) 30° (B) 45° (C) 60° (D) 90°

**Q4.** A vertical pole 6 m high casts a shadow $2\sqrt{3}$ m long on the level ground. The sun's elevation is: [1]

(A) 60° (B) 45° (C) 30° (D) 90°

**Q5. Assertion (A):** If the length of the shadow of a vertical pole is equal to its height, then the angle of elevation of the sun is 45°.
**Reason (R):** For any right-angled triangle, when the opposite side equals the adjacent side, $\tan\theta = 1$, which gives $\theta = 45°$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

## SECTION B: Short Answer Questions (11 Marks)

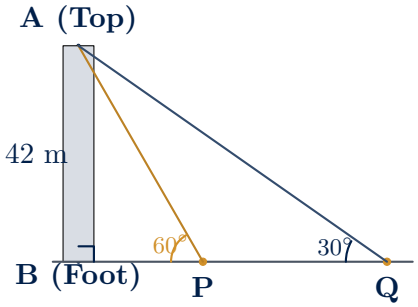
**Q6.** A vertical pole stands on level ground. From a point on the ground 20 m away from the foot of the pole, the angle of elevation of the top is 60°. Find the height of the pole. [2]

**Q7.** As observed from the top of a 75 m high lighthouse from the sea level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships. [Take $\sqrt{3} = 1.732$] [3]

**Q8.** Two poles of equal heights are standing opposite each other on either side of the road, which is 80 m wide. From a point between them on the road, the angles of elevation of the top of the poles are 60° and 30° respectively. Find the height of the poles and the distances of the point from the poles. [3]

**Q9.** From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. Determine the height of the tower. [3]

## Q10. CASE STUDY: The Historic India Gate Monument (New Delhi)




India Gate is a triumphal arch war memorial located astride the Rajpath in New Delhi. It stands at a majestic height of 42 m. During a mathematics educational field trip, students of Class 10 visited the monument to measure angles and compute distances using trigonometric principles.
A student standing at point $P$ on the ground measures the angle of elevation of the top of India Gate ($A$) as $60°$. Another student stands further away at point $Q$ in the same vertical plane and observes the top at an angle of elevation of $30°$. The foot of the monument is represented by point $B$.

Based on the above mathematical scenario, answer the following questions:

(i) At 12 noon, when the sun's rays are inclined at $45°$ to the horizontal ground, calculate the length of the shadow cast by India Gate. **[1]**

(ii) Find the distance of student $P$ from the foot of the monument $B$. **[1]**

(iii) Find the distance between the two students (i.e., length $PQ$). **[2]**

**— OR —**

Find the ratio of the length of the shadow cast by India Gate when the sun's altitude is $30°$ to that when the sun's altitude is $60°$. **[2]**

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★